SANKET · AI — EXECUTIVE JURY BRIEF

Smart India Hackathon 2026 · Problem Statement SIH26165

Ministry of Petroleum & Natural Gas · Oil India Limited (Upper Assam Basin)

AI/NLP Engine for SIF Precursor Detection & Cross-Shift Bow-Tie Intelligence

1. The Human Anchor: Baghjan Well #5 (May 27, 2020)

During the Baghjan blowout in Tinsukia, Assam (burning 160+ days, 3 firefighter fatalities, ■580 Cr loss), the DGMS inquiry proved the disaster was **NOT an unpredictable act of God**. For 14 consecutive days, precursors were noticed across shifts: mud loss (Day -14), casing vibration (Day -10), cellar gas bubbling (Day -6), and a leaking BOP valve (Day -3). **Because Shift B never saw Shift A's paper logbook, the signals remained disconnected until the explosion.**

2. SANKET AI Architectural USPs (Why We Win)

Core Capability	Industry Reality / Existing Gaps	SANKET · AI Proprietary Solution
Indic Multilingual Voice	Roughnecks speak Assamese/Hindi; typed English forms are ignored or filled 4 days late.	Zero-tap voice notes in Assamese, Hindi, Bengali & English with automatic acoustic entity extraction.
Bow-Tie Causal Graph	Isolated rows in Excel; no link between a valve leak today and a pressure surge tomorrow.	NetworkX graph correlates multi-day signals into active degradation chains before the blowout apex.
Air-Gapped Sovereign	Cloud-dependent solutions fail at remote drilling wellpads with zero 4G/satellite link.	100% offline CPU embeddings (multilingual-e5-small) & SQLite. ■0 monthly cloud API dependency.
Sub-Second SWA Reflex	Paper PTW approval takes hours; stop-work authority is rarely exercised in real-time.	Instant Emergency Stop Work trigger locks DCS perimeter, alerts Control Room sirens & creates SHA-256 audit.

3. Live Demonstration Flight Plan (4 Killer Screens)

Screen & Route	What Judges See	Strategic Impact / Regulatory Compliance
Control Room Siren /dashboard/control-room	Live DCS telemetry (Wellhead PSI, H2S PPM, Blowout Barrier integrity, audio sirens).	Standard OISD-189 Real-Time Process Barrier Monitoring.
Baghjan Disaster Replay /dashboard/replay	Chronological time-travel through the 14 days of May 2020 warning signals.	Demonstrates how SANKET would have halted the blowout on Day -6.
Bow-Tie Safety Graph /dashboard/graph	Interactive causal network linking threats, barriers, precursors, and consequence apexes.	Proves cross-shift intelligence correlating isolated shift reports.
Mobile Worker Terminal /worker	Ruggedized frontline phone UI: 1-Tap Assamese Voice, Quick Incidents, Red SWA Button.	Empowers roughnecks with statutory Stop Work Authority without friction.

Statutory Alignment: Directorate General of Mines Safety (DGMS) Guidelines · Oil Industry Safety Directorate **OISD-189 / OISD-GDN-206** · IOGP Report 459 (9 Life-Saving Rules) · ISO 45001 / OSHA 1910 Process Safety Management.

Live Cloud URL: https://oil-sanket.up.railway.app · **Scan the QR Code at the top-right to test the live mobile frontline worker terminal directly on your phone.**